

# EECS 2311: CRM Planning Map

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## Vision Statement

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ClubRM is a simple Customer Relationship Management (CRM) tool built for student clubs. It helps organizations organize their contacts, track their history with professionals, and stop losing valuable network data when student leaders graduate.

Student clubs are the bridge between students and the professional world (sponsors, mentors, speakers). But clubs have a major flaw. Currently, valuable relationships are "owned" by individual students, not the club. When a President or VP graduates, their personal connections and email history disappear with them. The new team has to start from scratch every September, often awkwardly introducing themselves to professionals who have already worked with the club. Important contact info is currently messy- scattered across random spreadsheets, personal messages, and lost email threads.

We want student clubs to operate with the organization and memory of a real business. ClubRM serves as a central vault for a club's network. ClubRM allows a club to look up a contact like a recruiter or a guest speaker, and instantly see every interaction the club has ever had with them. This means when a new executive team takes over, they aren't looking at a blank page, they are building on the previous team's work. We are replacing the chaos of spreadsheets with a clean, easy-to-use tool that tracks who people are and what they have done for the club.

Our primary users are student club leaders. This includes Presidents, VPs, and anyone responsible for finding sponsors or speakers. The secondary beneficiaries are future club members, who will inherit an organized history instead of a complete mess.

This is useful for several reasons. The relationship stays with the club, even when the student leaves. It stops clubs from accidentally emailing the same sponsor twice or forgetting past promises, making the club look more professional. It also saves time and energy. Student volunteers can focus on planning events rather than hunting for lost email addresses.

For this project to be successful, the Minimum Viable Product must be able to do several things. Users should be able to easily add, view, edit, and delete contact profiles. A timeline feature where users can log meetings, emails, and notes for each contact would be useful. They should be able to quickly find people based on labels (e.g., "Sponsor," "Alumni," "Speaker"). One club shouldn't see

another club's data. (ie, Club A's sponsors are private from Club B). Finally, the interface must be simple enough that a student leader can figure it out immediately without a manual/walkthrough.

## Big User Story (Taziz)

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Centralized Contact and Organization Management

As a Club Executive (VP/President), I want to maintain a centralized, persistent database of industry professionals and organizations so that the club's network is preserved independently of the current leadership team, preventing data loss when executives graduate. This includes the ability to categorize contacts (sponsors, speakers, mentors) and view their affiliation details.

Priority: High

Cost: 2 weeks

## Big User Story (Yusuf)

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Confidentiality with CRM access

As a VP of Communications, I would like ClubRM to ensure that only authorized personnel are able to make use of ClubRM's functions to ensure that there is a certain level of privacy and authority when it comes to maintaining contacts. Club executives, like myself, would like the CRM to be used in order to maintain good standards relations with clients on a personal level so that there is that ease with communication. In order for there to be ease with communication, the pool of those authorized to do so needs to be small so as not to overwhelm the clients.

Priority: Medium

Cost: 1 week

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## Iteration 1 User Stories

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### User Story (Taziz)

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## Add New Professional Contact

As a VP of External Affairs, I want to manually input a new professional's details (Name, Job Title, Organization, Email, LinkedIn URL) into the system so that I can quickly have a digital record of a contact I made at an event before I lose their business card or forget the context. The form must collect: Full Name, Email, Organization, Job Title, and LinkedIn URL. The system must prevent submission if the email format is invalid (e.g., missing "@"). I should be able to assign a tag (e.g., "Potential Speaker") during creation. Clicking save closes it, a "Success" popup appears, and the new contact is visible at the top of the contact list.

Priority: High

Cost: 2 days

## User Story (Yusuf)

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### User Login Configuration

As a Club Coordinator, I would like to have a fully featured Login Screen that would handle Account creation well through the "Sign Up" button. Afterwards, I would like the Login Screen to have features like the "Show Password" feature that would make it easier for me to remember what password I am inputting as well as the required length. I would also like for there to be a "Forgot Password" link so that I would be able to change up my password however I would like. A standard password for security purposes would need to have at least 1 special character, 1 capital letter, length greater than or equal to 8, and numerical characters as well.

Priority: High

Cost: 3 days

### User Story (Daniel) Sophisticated Sorting

As a Club Executive (VP/President), I would like to be able to view emails, contact profiles, and notes of my club on a central hub each separated into their own tabs. On each tab, I can Sort by Date, and Sort by Name, as well as both in reverse. Additionally, I want to be able to (optionally) Group these categories in a similar manner. On each entry, the Date Modified should be displayed. Furthermore, I should be able to further subdivide each tab by labels currently assigned, add more, or edit the ones that are currently listed. This submenu should list the pre-existing labels already in-use, allowing for easier consistency. It's important that the system remembers my configuration of the information when I log off and return to it in the future.

Priority: High

Cost: 4 days

### **User Story (Ashdeep)** View Professional Contact Profile

As a Club Executive (VP/President), I want to view the full profile of a professional contact from the contact list so that I can quickly understand who they are, their role, and their relationship with the club before reaching out or planning an event.

When I click on a contact from the contact list, I should be able to see their Full Name, Job Title, Organization, Email, LinkedIn URL, assigned tags, and the date the contact was added. The information should be displayed clearly on a single page or panel. I should be able to return to the contact list without losing my current sorting or filtering settings.

Priority: High Cost: 1.5 days

## **User Story (Shivam)**

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### Edit Existing Professional Contact

As a VP of External Affairs, I want to edit the details of an existing professional contact so that I can keep their information accurate if their role, organization, or contact details change.

From a professional contact's profile, I should be able to update fields such as Job Title, Organization, Email, LinkedIn URL, and assigned tags. The system must check the email format before allowing changes to be saved. After saving, a confirmation message should appear, and the updated information should be immediately visible in the contact profile and contact list.

Priority: Medium Cost: 2 days

## **Iteration 2 and 3 User Stories**

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### **User Story (Yusuf)**

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#### Automated Document Categorization

Developer: Yusuf Garba Task: Update the DocumentsPage to include a category dropdown and map it to a category column in the Supabase documents table Est. Time: 3 days Actual Time: 3 days

Status: Completed

## User Story (Yusuf)

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Document Version Tracking

Developer: Yusuf Garba Task: Implement a History side-panel GUI to display a modification log for specific files ordered by update metadata Est. Time: 2 days Actual Time: 2 days Status: Completed

## User Story (Shivam)

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Member Directory Export

Developer: Shivam Patel Task: Create a utility function and GUI button to convert the JSON result from the members query into a downloadable CSV blob Est. Time: 2 days Actual Time: 2 days Status: Completed

## User Story (Daniel)

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Role-Based Access Control (RBAC) UI

Developer: Daniel Kyere Task: Create an "Access Management" GUI in Settings to update user roles and permissions within the profile table Est. Time: 4 days Actual Time: 4 days Status: Completed

## User Story (Daniel)

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Security Audit Log Viewer

Developer: Daniel Kyere Task: Implement a "logs" table and GUI fetcher to display the most recent sensitive system actions in a paginated list Est. Time: 3 days Actual Time: 3 days Status: Completed

## User Story (Ashdeep)

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First-Time User Guided Tour

Developer: Ashdeep Singh Task: Implement an interactive tooltip-based tour for the Sidebar and Finance pages triggered by a database status flag Est. Time: 3 days Actual Time: 3 days Status: Completed

## User Story (Ashdeep)

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Organization Setup Portal

Developer: Ashdeep Singh Task: Build a GUI screen for users to create or join organizations via invite codes linked to the Supabase backend Est. Time: 4 days Actual Time: 5 days Status: Completed

## User Story (Naeem)

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Event & Calendar Management

Developer: Naeem Baig Task: Build the Events Page with Month/Week/Day calendar views, event creation/editing modal, event detail panel, and events context provider Est. Time: 4 days Actual Time: 3 days Status: Completed

## User Story (Naeem)

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Multi-View Task Board

Developer: Naeem Baig Task: Build the Tasks Page with Board (Kanban drag-and-drop), List (sortable/groupable), Timeline (Gantt chart), Calendar, and Workflow views, plus task modal and tasks context provider Est. Time: 5 days Actual Time: 5 days Status: Completed

## User Story (Naeem)

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Club Finance Management

Developer: Naeem Baig Task: Build the Finance Page with Budget Overview (charts, stats), Expenses tab (CRUD, approve/deny), Reimbursements tab (submit/approve/deny/paid flow), Income tab (dues, sponsorships, fundraising), and Analytics tab (trends, burn rate, insights) Est. Time: 3 days Actual Time: 2 days Status: Completed

## User Story (Taziz)

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Customizable Dashboard Engine

Developer: Taziz Ahsan Task: Implement a persistent drag-and-drop layout system using @dnd-kit across all 8 role-based dashboards, including widget visibility management. Est. Time: 4 days Actual Time: 3 days Status: Completed

## User Story (Taziz)

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Live Data Dashboard Integration

Developer: Taziz Ahsan Task: Connect all role-specific dashboards to real-time application state via React Contexts and implement a secure JWT-authenticated KPI Backend API. Est. Time: 5 days Actual Time: 4 days Status: Completed

## User Story (Taziz)

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Comprehensive Testing Infrastructure

Developer: Taziz Ahsan Task: Establish a robust testing suite comprising unit tests (with logic stubs) and integration tests (live Supabase connectivity) to ensure project stability. Est. Time: 3 days Actual Time: 2 days Status: Completed

## User Story (Taziz)

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Omniscient Global Search & Navigator

Developer: Taziz Ahsan Task: Implement cross-module indexing for all 6 primary modules, build keyboard-driven UX (Ctrl+K), and develop persistent search history engine. Est. Time: 4 days Actual Time: 4 days Status: Completed

## User Story (Taziz)

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End-to-End Professional Relationship CRM — Global CRM State & Directory Architecture

Developer: Taziz Ahsan Task: Architect global SponsorsContext provider for unified CRUD management Est. Time: 2 days Actual Time: 2 days Status: Completed

## User Story (Taziz)

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End-to-End Professional Relationship CRM — Contact Lifecycle & Visualization Sheets

Developer: Taziz Ahsan Task: Implement Contact Profile Visualization Sheet with LinkedIn integration Est. Time: 1 day Actual Time: 1 day Status: Completed

## User Story (Taziz)

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Production-Grade Reliability & Verification Framework

Developer: Taziz Ahsan Task: Develop comprehensive Unit & Integration suite for all 5 Context modules Est. Time: 3 days Actual Time: 3 days Status: Completed

## User Story (Ashdeep)

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Onboarding Flow

Developer: Ashdeep Singh Task: Design and implement a multi-step "Welcome" wizard for first-time users to improve user experience and authorization Est. Time: 3 days Actual Time: 3 days Status: In Progress

## User Story (Yusuf)

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Tracking Documents in Database

Developer: Yusuf Garba Task: Implement a Document Upload system using Supabase Storage buckets. Est. Time: 3 days Actual Time: 3 days Status: In Progress

# Iteration 3 Developer Stories

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## Developer Story (Yusuf)

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UI Decomposition

Developer: Yusuf Garba Task: Split DashboardPage into StatsCard and Activity components to resolve "Large Class" smell Est. Time: 6 hours Actual Time: 6 hours Status: Completed

## Developer Story (Naeem)

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Search Standardization

Developer: Naeem Baig Task: Implement SearchStrategy interface to unify search logic across Finance and Contacts modules Est. Time: 5 hours Actual Time: 8 hours Status: Completed

## Developer Story (Shivam)

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Data Layer Hardening

Developer: Shivam Patel Task: Replace primitive data types with Transaction and Event domain objects for validation Est. Time: 4 hours Actual Time: 4 hours Status: Completed

## Developer Story (Taziz)

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Logic Decoupling

Developer: Taziz Ahsan Task: Extract search and filter logic from ContactsPage.tsx into useContactSearch hook Est. Time: 4 hours Actual Time: 5 hours Status: Completed

# Fully Implemented User Stories with GUI

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## 1. Role-Based Dashboards Detail Stories:

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1. [Taziz Ahsan] 8 different custom dashboards that load automatically based on the user's role (like president, vp-internal, etc.).
2. [Naeem Baig] An "Approval Queue" widget that lets you approve finance requests right from the home page.
3. [Taziz Ahsan] An "Org Health" and "Risk Alert" widget that calculates a score based on club data.
4. [Taziz Ahsan] A drag-and-drop feature so users can rearrange and customize the widgets on their dashboard.

## 2. Global Search Menu Detail Stories:

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1. [Taziz Ahsan] A global search bar that opens anytime you press Cmd+K anywhere on the site.
2. [Taziz Ahsan] The search results instantly filter across all models at once (Events, Tasks, Members, Sponsors, Documents, Finance).
3. [Taziz Ahsan] The search menu remembers your history and shows your "Recent Searches" the next time you open it.

### **3. Reports & Analytics Page Detail Stories:**

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1. [Yusuf Garba] A main dashboard page that combines data from members, operations, finances, and events.
2. [Taziz Ahsan] Visual charts (like pie charts and radial dials) that show member retention and task completion rates.
3. [Yusuf Garba] Graphs showing monthly club activity and spending trends over time.
4. [Taziz Ahsan] A "Download CSV" button that grabs all this data and turns it into a downloadable spreadsheet file.

### **4. Sponsorship Pipeline Detail Stories:**

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1. [Daniel Kyere] A management page to organize corporate sponsors by tiers (Platinum, Gold, etc.).
2. [Daniel Kyere] A pipeline view to track where sponsors are in the process (Prospect, Pending, Active, Churned).
3. [Ashdeep Singh] A history log to keep track of every email, call, or meeting with each sponsor contact.

### **5. Club Finance Management Detail Stories:**

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1. [Naeem Baig] A budget overview panel showing how much money is left using progress bars.
2. [Naeem Baig] A table to view, add, and edit expenses, with clickable column headers for sorting.
3. [Naeem Baig] A pie chart section to visually break down the sources of club income.
4. [Naeem Baig] A workflow for handling member reimbursements from "Pending" to "Approved" to "Paid".

### **6. Marketing & Social Media Hub Detail Stories:**

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1. [Shivam Patel] The marketing page includes a dedicated Analytics tab featuring a bar chart comparing reach vs. engagement per campaign.
  2. [Shivam Patel] A live feed simulation showing social media posts with "likes" and "shares".
  3. [Shivam Patel] Clicking into any campaign opens a detailed dialog showing a full budget usage progress bar, a breakdown of every post within that campaign.
  4. [Shivam Patel] A personal profile page where users can manage their info and link their social media accounts including Instagram, Facebook, LinkedIn, X/Twitter, and TikTok, with all data persisted to a Supabase database and clickable preview links that open each profile directly.

## **7. Event & Calendar Management Detail Stories:**

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1. [Naeem Baig] A large monthly calendar that displays all upcoming events.
2. [Naeem Baig] A popup modal to create new events and pick a custom color and tag for them.
3. [Naeem Baig] An event details view that shows the maximum capacity and how many people have RSVP'd.
4. [Shivam Patel] A popup button that allows events to be synced onto a Google Calendar for personal use.

## **8. Task Board & Timelines Detail Stories:**

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1. [Yusuf Garba] A Kanban board where you can drag and drop tasks between different status columns.
2. [Yusuf Garba] A timeline (Gantt chart) view that plots tasks on a calendar grid based on their dates.
3. [Yusuf Garba] Visual dashed lines (drawn using SVGs) that automatically connect parent tasks to their dependent child tasks on the timeline.

## **9. Member Directory Detail Stories:**

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1. [Ashdeep Singh] A full roster grid showing all club members, which can be filtered by role and status.
2. [Ashdeep Singh] Profile cards that show a member's total tasks completed and events attended.

- 3. [Taziz Ahsan] Automatically generated custom avatar colors based on the letters in a member's name.

## 10. Onboarding Flow Detail Stories:

- 1. [Ashdeep Singh] Designed a step-by-step onboarding process for new users, where they enter their basic details before being allowed into the main app.
- 2. [Ashdeep Singh] Implemented profile setup rules to ensure users cannot choose restricted roles (like admin-level roles) during onboarding.
- 3. [Ashdeep Singh] Added logic to control access so that users must complete onboarding before they can access the main application features.

## Testing Distribution

To make sure our testing is unbiased, we have distributed the stories among the group members. No team member is assigned to test a feature that they were heavily involved in developing.

Team Members: Yusuf Garba, Shivam Patel, Taziz Ahsan, Ashdeep Singh, Daniel Kyere, Naeem Baig

| Team Member   | Assigned Test Feature 1             | Assigned Test Feature 2            |
|---------------|-------------------------------------|------------------------------------|
| Yusuf Garba   | Feature 5: Finance & Reimbursements | Feature 7: Events & Calendars      |
| Shivam Patel  | Feature 4: Sponsorship Pipeline     | Feature 9: Member Directory        |
| Taziz Ahsan   | Feature 1: Role-Based Dashboards    | Feature 8: Task Timelines          |
| Ashdeep Singh | Feature 2: Global Search Menu       | Feature 6: Marketing Hub           |
| Daniel Kyere  | Feature 10: Onboarding Flow         | Feature 3: Reports & Analytics     |
| Naeem Baig    | Feature 4: External Sponsor History | Feature 3: Analytics CSV Downloads |

## Planning Map

| Detailed User Story                             | Who it's assigned to                                |
|-------------------------------------------------|-----------------------------------------------------|
| Add New Professional Contact                    | Taziz Ahsan                                         |
| View Professional Contact Profile               | Ashdeep Singh                                       |
| Sophisticated Sorting                           | Naeem Baig, Yusuf Garba, Ashdeep Singh, Taziz Ahsan |
| User Login Configuration                        | Daniel Kyere                                        |
| Edit Existing Professional Contact              | Shivam Patel                                        |
| Centralized Contact and Organization Management | Taziz Ahsan                                         |
| Automated Document Categorization               | Yusuf Garba                                         |
| Document Version Tracking                       | Yusuf Garba                                         |
| Member Directory Export                         | Shivam Patel                                        |
| Live Profile Preview Cards                      | Shivam Patel                                        |
| Role-Based Access Control (RBAC) UI             | Daniel Kyere                                        |
| Security Audit Log Viewer                       | Daniel Kyere                                        |
| First-Time User Guided Tour                     | Ashdeep Singh                                       |
| Organization Setup Portal                       | Ashdeep Singh                                       |
| Event & Calendar Management                     | Naeem Baig                                          |
| Multi-View Task Board                           | Naeem Baig                                          |
| Club Finance Management                         | Naeem Baig                                          |
| Customizable Dashboard Engine                   | Taziz Ahsan                                         |
| Live Data Dashboard Integration                 | Taziz Ahsan                                         |
| Comprehensive Testing Infrastructure            | Taziz Ahsan                                         |
| Omniscient Global Search & Navigator            | Taziz Ahsan                                         |

| Detailed User Story                                          | Who it's assigned to |
|--------------------------------------------------------------|----------------------|
| End-to-End Professional Relationship CRM — Global CRM State  | Taziz Ahsan          |
| End-to-End Professional Relationship CRM — Contact Lifecycle | Taziz Ahsan          |
| Production-Grade Reliability & Verification Framework        | Taziz Ahsan          |
| Onboarding Flow                                              | Ashdeep Singh        |
| Social Media & External Profile Integration                  | Shivam Patel         |
| Tracking Documents in Database                               | Yusuf Garba          |
| Confidentiality with CRM Access                              | Daniel Kyere         |